

# KEVIN KAKOLLA

Senior ML Engineer · GenAI/LLM · Full-Stack · Cloud

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## PROFESSIONAL SUMMARY

Senior ML/GenAI Engineer with 15+ years across financial services, energy, healthcare, and media sectors. Delivers end-to-end AI solutions — from data pipelines and model training to production deployment — using Python, PyTorch, LLMs, RAG, LangChain, Azure ML, JavaScript, React/TypeScript. Currently completing a Graduate Certificate in Generative AI & LLM at Carnegie Mellon University, with active research on multimodal web agents using DeBERTa and GPT-4o.

## TECHNICAL SKILLS

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|----------------------|------------------------------------------------------------------------------------------|
| <b>ML / AI</b>       | PyTorch, HuggingFace, DeBERTa, scikit-learn, XGBoost, TensorFlow, Keras, CNNs, RNNs      |
| <b>LLM / GenAI</b>   | GPT-4o, Claude, Gemini, Flan-T5, LangChain, LangGraph, RAG, Pinecone, Autogen            |
| <b>Cloud / MLOps</b> | Azure ML, Azure OpenAI, Databricks, AWS SageMaker, Vertex AI, MLflow, Airflow, Terraform |
| <b>Backend</b>       | Python, Django, FastAPI, Node.js, Java, Spring Boot, PostgreSQL, MongoDB, Oracle         |
| <b>Frontend</b>      | React, TypeScript, Redux, Angular, Material UI, Tailwind, HTML5, CSS3                    |
| <b>DevOps</b>        | Docker, Kubernetes (AKS), Azure DevOps, GitHub Actions, Jenkins, CI/CD pipelines         |
| <b>Data</b>          | Pandas, NumPy, PySpark, Dask, BigQuery, Azure Synapse, SQL Server, Matplotlib, Seaborn   |

## EDUCATION

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|--------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Graduate Certificate — Generative AI &amp; LLM</b>   <a href="#">Carnegie Mellon University</a>           | <i>Jan 2026 – Dec 2026</i> |
| <b>Data Science &amp; Machine Learning</b>   <a href="#">University of Toronto — Data Sciences Institute</a> | <i>Mar – Aug 2025</i>      |
| <b>AI &amp; Data Science Program</b>   <a href="#">UC Berkeley</a>                                           | <i>Fall 2024</i>           |
| <b>Foundations of Large Language Models</b>   <a href="#">University of Waterloo</a>                         | <i>2024</i>                |
| <b>Artificial Intelligence (ML, DL, RL)</b>   <a href="#">University of Toronto</a>                          | <i>2023 – 2024</i>         |
| <b>MBA — Business Intelligence &amp; Data Analytics</b>   <a href="#">University of Calgary</a>              | <i>May 2014</i>            |
| <b>B.Eng. — Computer Science</b>   <a href="#">S.V.U. College of Engineering, India</a>                      | <i>2000</i>                |

## PROFESSIONAL EXPERIENCE

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| <b>GenAI/LLM Engineer</b>   <a href="#">Fleetworthy</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>Oct 2025 – Apr, 2026</i> |
| <ul style="list-style-type: none"><li>▶ Building Llama 4 Sales voice agent using fine-tuned Llama 4 Maverick 17 Billion params, 128 experts, total 400B params using Mixture of Experts (MoE)</li><li>▶ Developing Microservices for SmartHub CRM app for AI Agentic Sales Voice agent using FastAPI</li><li>▶ Supervised Fine Tuning of LLM models based on customer and sales data using QLoRA (Quantized Low Ranking Adaptation), DPO (Direct Preference Optimization)</li><li>▶ Eval using LLM (GPT-5) as Judge for LLM response</li><li>▶ Quantization using QLoRA, DPO, vLLM, LLM-as-a-Judge, Locust 1000 users</li><li>▶ Trained <b>DeBERTa-v3-base</b> candidate generation model on Multimodal-Mind2Web dataset (7,775 examples) for DOM element ranking in language-guided browser automation tasks</li><li>▶ Built two-stage web agent pipeline: DeBERTa candidate ranker + <b>GPT-4o</b> action prediction, integrated with Playwright for end-to-end browser task execution</li></ul> |                             |

- ▶ Applied RAG, LangChain, and LangGraph to adapt LLM reasoning for fleet-specific domain workflows; evaluated using MRR, element accuracy, and action F1
- ▶ Built Django/PostgreSQL backend APIs for fleet inspection, user management, and vehicle lifecycle workflows at Fleetworthy

#### Machine Learning Engineer | [T6 Health Systems](#)

May 2025 – Sept 2025

- ▶ Designed ML pipeline for **radiation exposure risk prediction** using XGBoost, ridge/lasso regression on biomedical and environmental sensor data; SHAP-based explainability for clinical interpretability
- ▶ Applied ResNet50 and VGG16 for biomedical image classification; integrated HALO AI tools for whole slide imaging and clinical decision support
- ▶ Built React/TypeScript real-time dashboards for health monitoring in mobile and desktop environments

#### Machine Learning Engineer | [Ontario Power Generation](#)

Jun 2023 – Mar 2025

- ▶ Built **enterprise GenAI platform** on Azure (Databricks, ML Studio, Azure OpenAI, Cognitive Search) for intelligent document search, automated report generation, and predictive maintenance
- ▶ Trained and deployed ML models for anomaly detection, image-based inspection (VGG16/ResNet50), and radiation permit confidence scoring in regulated environments
- ▶ Deployed containerized ML services with full CI/CD via Azure DevOps, Terraform, Docker, and AKS; built scalable AI pipelines on Azure Synapse and SQL Server

#### Machine Learning Engineer | [Scotia Bank](#)

Jan 2023 – Jun 2023

- ▶ Built credit card fraud detection pipeline using K-NN, logistic regression, and SVM; fine-tuned OpenAI LLM with zero-shot/few-shot learning for NLP classification tasks
- ▶ Designed FastAPI microservices with authentication and monitoring; automated CI/CD model deployment pipelines on Azure DevOps

#### Machine Learning Engineer | [Bell Canada](#)

Jul 2021 – Jan 2023

- ▶ Developed NLP classification, object detection, and customer churn prediction pipelines using scikit-learn and sentence transformers; deployed model training toolchains end-to-end
- ▶ Built Wi-Fi checkup and video feed features in React/Redux deployed to production for desktop and mobile

#### ML Engineer | [Goodfood/](#)

an 2021 – Jun 2021

- ▶ Built food recommendation system using clustering/classification ML algorithms; deployed via MLflow and Vertex AI; implemented CNN-based recipe and grocery image classification
- ▶ Sentiment analysis using HuggingFace and OpenAI for recipe reviews; price prediction models optimized with MAE, F1, and accuracy metrics

#### Machine Learning Engineer | [Bell Media](#)

Sept 2019 – Dec 2020

- ▶ Built movie/TV sentiment analysis with OpenAI and HuggingFace; show recommendation system using clustering algorithms; video genre classification with softmax CNN in Python
- ▶ Developed Crave/CTV media application in ReactJS with Apollo GraphQL; data analysis on BigQuery, model deployment via Vertex AI

#### ML Engineer — Oncology & Biomedical Imaging | [CBRE / Navigating Cancer/](#)

an 2019 – Oct 2019

- ▶ Predicted patient appointment no-shows, Perjeta drug risk scores, and high-risk patient triage using ML models; integrated HALO AI tools (Breast IHC, Lung PD-L1, Prostate AI) for whole slide image analysis
- ▶ Built Django/Flask backend for clinical decision support; collaborated with oncologists on pathology-inspired datasets

#### **Angular/React Developer** | [TD Bank](#)

*Aug 2018 – Jan 2019*

- ▶ Built TD RESL Home Journey Mortgage Application; Salesforce backend integration; CI/CD with Docker and Jenkins; accessibility testing with JAWS and VoiceOver

#### **React/Angular Developer** | [Equifax](#)

*Mar 2018 – Aug 2018*

- ▶ Deployed Interconnect App on AWS using Apollo GraphQL; Spring Boot microservices with Java 8, Eureka server; CI/CD via Docker, Jenkins, Bitbucket

#### **Angular Developer** | [Scotia Bank](#)

*Jul 2017 – Feb 2018*

- ▶ Built Zeus Trade Management BackOffice UI for MIFID II (FIX Trading) using Angular/React/Redux; stock prediction using linear regression in scikit-learn

#### **Angular UI Developer** | [Sci MarketView](#)

*Jan 2017 – Jun 2017*

- ▶ Responsive Driver Application for automobile dealers using AngularJS/Angular2/React; TypeScript, reactive forms, Bootstrap; desktop, Android, and iPhone support

#### **Java Developer** | [Royal Bank of Canada](#)

*Aug 2016 – Dec 2016*

- ▶ UI Client & Advisor portal using AngularJS 1.5, NodeJS, Jasmine/Karma/Protractor testing; deployed on IBM Bluemix, AWS, and Pivotal Cloud Foundry

#### **Java Developer** | [TD Bank / Rogers](#)

*Aug 2015 – Aug 2016*

- ▶ Built TD SimpleApp form generation and Rogers Sales & Service Portal using AngularJS, jQuery, Bootstrap, ATG/WebLogic

#### **Project Manager / Java Developer** | [Southern Alberta Institute of Technology](#)

*Jun 2014 – Aug 2015*

- ▶ MVC implementation using Struts/Servlet/JSP for student registration system; jQuery/AngularJS UI; Bootstrap responsive design

*Earlier roles: Polycom/Tata Elxsi (Java/J2EE, 2012), JPMorgan Chase/Dish Network/Union Pacific (EJB/JSF/Hibernate, 2006–2011), Infosys Technologies (EJB/WebLogic, 2004–2006), Hewlett Packard Riyadh (EAI integration, 2003–2004), Fujitsu Network Communications San Jose — implemented RIPv1/v2, OSPF, IS-IS routing protocols in C++ (2000–2002).*